

VARDHAN BURANDE

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PROFILE

Data professional with hands-on experience building ETL pipelines, analytical models, and dashboards across energy, manufacturing, aviation, e-commerce, and consumer data. Combines Python, SQL, Power BI, and machine learning with strong data-quality discipline and clear stakeholder communication.

Core expertise: ETL & data quality · SQL & statistical analysis · Machine learning · Power BI & Streamlit · Time-series & sensor data

TECHNICAL SKILLS

Languages: Python (Pandas, NumPy, Scikit-learn, Plotly, Matplotlib, Seaborn), R, SQL

Databases: MySQL, SQLite, SQL Server (window functions, CTEs, subqueries, joins, aggregations)

Visualization: Power BI (DAX, data modeling, drill-through), Streamlit, Excel (PivotTables, VLOOKUP)

ML / Stats: Regression, Classification, Time-Series Forecasting, EDA, Hypothesis Testing, Feature Engineering, LightGBM, MiniRocket

Tools: Git/GitHub, Jupyter, VS Code, ETL pipeline design, data cleaning, model deployment

EXPERIENCE

Research Assistant · *University at Buffalo, Visual Computing Lab*

Mar 2026 – Present | Buffalo, NY

- Built Python pipelines to clean and structure NYISO electricity-market data, map generator records to nodal locations, and prepare hourly datasets for day-ahead and real-time price analysis.
- Analyzed price, congestion, and weather data using seasonal correlation analysis and outlier detection to identify factors associated with price spikes across high-impact nodes.
- Created visualizations and research summaries that explained price variation, seasonal demand patterns, and potential Virtual Power Plant opportunities to faculty and project stakeholders.

Data Engineer Intern · *Machinery Monitoring Systems, LLC*

Aug 2025 – Jan 2026 | Buffalo, NY

- Built an end-to-end ETL pipeline that converted raw tri-axial vibration data into model-ready datasets using signal filtering, normalization, and automated quality checks.
- Processed and labeled 21,000+ vibration samples across four fault classes, resolved data-quality issues, and created stratified train-test splits for reliable model evaluation.
- Trained and packaged a MiniRocket and Ridge Classifier model that achieved 99.96% accuracy with a 1.7 MB footprint, supporting edge-based equipment monitoring.

KEY PROJECTS

Smart Workout Tracker: ML-Powered Analytics App [\[Live\]](#) [\[GitHub\]](#)

- Built an ETL pipeline for IMU sensor data covering ingestion, outlier handling, normalization, and feature engineering; trained a classifier that achieved 99.25% accuracy on 929 test samples.
- Deployed a Streamlit app with session analytics, trend charts, multi-file upload, and CSV export, allowing users to review and compare workout data in one place.

Fly-Trend Hub: Airline Operations Analytics Dashboard [\[Live\]](#) [\[GitHub\]](#)

- Designed a normalized SQLite database for airline operations data and wrote SQL queries using CTEs, joins, and aggregations to support dashboard reporting.
- Combined multiple airline datasets in Pandas and built a Streamlit dashboard that tracked flight status, passenger trends, and monthly departure patterns.

E-commerce Revenue & Retention Analysis [\[GitHub\]](#)

- Identified a 3.12% repeat-customer rate and approximately \$97K in revenue lost to cancellations through analysis in SQL Server, Power BI, and Python.
- Built RFM segments and cohort-retention views to identify at-risk customer groups and support targeted retention decisions.

House Price Prediction: Regression & Inference App [\[Live\]](#) [\[GitHub\]](#)

- Built a preprocessing pipeline for missing values, categorical encoding, and scaling, then tuned a LightGBM regression model through cross-validation.
- Deployed a Streamlit app with interactive EDA and real-time price estimates, allowing users to explore data and generate predictions.

EDUCATION

M.P.S. in Data Science and Applications · *University at Buffalo (SUNY)*

Aug 2024 – Dec 2025

GPA: 3.93/4.0 · Coursework: Machine Learning, Data Mining, Statistical Analysis, Database Systems, Data Visualization

B.E. in Mechanical Engineering · *SIES Graduate School of Technology*

Aug 2019 – May 2023

GPA: 3.81/4.0 · Mumbai, India